

# Measures

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*Remark.* A measure assigns a size to each set in a collection of sets: how many elements it has, how long it is, or how likely it is. This page builds measures from  $\sigma$ -algebras and additivity, using the [sets and functions](#) page's definitions. The Morrison Lab's probability notes define a probability measure as a measure that gives the whole space size 1 ([probability measure](#)).

## $\sigma$ -algebras

**Definition 0.1** ( $\sigma$ -algebra). A  $\sigma$ -algebra on a set  $S$  is a collection  $\mathcal{S}$  of [subsets](#) of  $S$  that satisfies:

- $\mathcal{S}$  contains  $S$  itself.
- For each set  $A$  in  $\mathcal{S}$ ,  $\mathcal{S}$  contains its complement  $S \setminus A$ .
- For each sequence  $A_1, A_2, \dots$  of sets in  $\mathcal{S}$ ,  $\mathcal{S}$  contains their union  $\bigcup_{i=1}^{\infty} A_i$ .

*Remark.* Other sources call it a  $\sigma$ -field (see [Wikipedia:  \$\sigma\$ -algebra](#)).

**Example 0.1** ( $\sigma$ -algebras for die rolls). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  ([sets of die rolls](#)), the collection of all subsets of  $D$  is a  $\sigma$ -algebra on  $D$ , since complements and unions of subsets of  $D$  are again subsets of  $D$ . So is the smaller collection

$$\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$$

which contains  $D$ , the complement of each of its sets, and every union of its sets: for example,  $\{2, 4, 6\} \cup \{1, 3, 5\} = D$ .

**Theorem 0.1** (Closure properties of a  $\sigma$ -algebra). *If  $\mathcal{S}$  is a  $\sigma$ -algebra on a set  $S$ , then  $\mathcal{S}$  contains:*

- the *empty set*  $\emptyset$ ;
- the union  $A_1 \cup \dots \cup A_n$  of any finitely many sets  $A_1, \dots, A_n$  in  $\mathcal{S}$ ;
- the intersection  $\bigcap_{i=1}^{\infty} A_i$  of any sequence  $A_1, A_2, \dots$  of sets in  $\mathcal{S}$ .

### **i** Proof

*Proof. Empty set.*  $\mathcal{S}$  contains  $S$ , so it contains the complement of  $S$ , which is  $S \setminus S = \emptyset$ .  
*Finite unions.* Extend  $A_1, \dots, A_n$  to a sequence by setting  $A_{n+1} = A_{n+2} = \dots = \emptyset$ ; each of these sets is in  $\mathcal{S}$ , by the first part. Then:

$$\begin{aligned} A_1 \cup \dots \cup A_n &= A_1 \cup \dots \cup A_n \cup \emptyset \cup \emptyset \cup \dots \quad (\text{a union with } \emptyset \text{ adds no elements}) \\ &= \bigcup_{i=1}^{\infty} A_i \quad (A_i = \emptyset \text{ for } i > n) \end{aligned}$$

and  $\mathcal{S}$  contains  $\bigcup_{i=1}^{\infty} A_i$  by the union rule of Definition 0.1.

*Countable intersections.* An element of  $S$  is in every  $A_i$  exactly when it is in none of the complements  $S \setminus A_i$ , so:

$$\bigcap_{i=1}^{\infty} A_i = S \setminus \bigcup_{i=1}^{\infty} (S \setminus A_i) \quad (\text{De Morgan's law})$$

Each  $S \setminus A_i$  is in  $\mathcal{S}$  by the complement rule, so their union is in  $\mathcal{S}$  by the union rule, and the complement of that union is in  $\mathcal{S}$  by the complement rule again.  $\square$

**Theorem 0.2** (All subsets form a  $\sigma$ -algebra). *For any set  $S$ , the collection of all subsets of  $S$  is a  $\sigma$ -algebra on  $S$ .*

### **i** Proof

*Proof.* Each condition of Definition 0.1 holds:

- $S$  is a subset of itself.
- For each subset  $A$  of  $S$ , the complement  $S \setminus A$  contains only elements of  $S$ , so it is a subset of  $S$ .
- For each sequence  $A_1, A_2, \dots$  of subsets of  $S$ , every element of  $\bigcup_{i=1}^{\infty} A_i$  is in some  $A_i$  and so in  $S$ ; the union is a subset of  $S$ .

$\square$

## Pairwise disjoint sets

**Definition 0.2** (Pairwise disjoint sets). Finitely or countably many sets  $A_1, A_2, \dots$  are **pairwise disjoint** (also called **disjoint** or **mutually disjoint**) when no two of them share an element:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

*Remark.* In probability, pairwise disjoint events are usually called **mutually exclusive**.

**Example 0.2** (Low and high die rolls). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  (sets of die rolls), the sets  $\{1, 2\}$  and  $\{5, 6\}$  are pairwise disjoint: no roll is in both. The sets  $\{2, 4, 6\}$  and  $\{1, 2\}$  are not, because 2 is in both.

## Additivity

**Definition 0.3** (Finite additivity). Let  $\mathcal{S}$  be a  $\sigma$ -algebra on a set  $S$ . A function  $\mu : \mathcal{S} \rightarrow [0, \infty]$ , with values in the extended non-negative reals, is **finitely additive** if, for every finite collection of pairwise disjoint sets  $A_1, \dots, A_n$  in  $\mathcal{S}$ , the value of their union is the sum of their values:

$$\mu(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mu(A_i)$$

**Example 0.3** (Counting elements is finitely additive). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  (sets of die rolls), let  $\mu(A) \stackrel{\text{def}}{=} |A|$ , the number of elements of  $A$ , for each set  $A$  in the  $\sigma$ -algebra of all subsets of  $D$  (Example 0.1). The sets  $\{1, 2\}$  and  $\{5, 6\}$  are pairwise disjoint (Example 0.2), and:

$$\begin{aligned} \mu(\{1, 2\} \cup \{5, 6\}) &= \mu(\{1, 2, 5, 6\}) && \text{(take the union)} \\ &= 4 && \text{(count the elements)} \\ &= 2 + 2 && \text{(write 4 as a sum)} \\ &= \mu(\{1, 2\}) + \mu(\{5, 6\}) && \text{(count each set's elements)} \end{aligned}$$

The same holds for any pairwise disjoint sets, because the sizes of disjoint sets add, so  $\mu$  is finitely additive.

**Definition 0.4** (Countable additivity). Let  $\mathcal{S}$  be a  $\sigma$ -algebra on a set  $S$ . A function  $\mu : \mathcal{S} \rightarrow [0, \infty]$  is **countably additive** (also called  $\sigma$ -additive) if, for every sequence of pairwise disjoint sets  $A_1, A_2, \dots$  in  $\mathcal{S}$ , the value of their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

**Example 0.4** (Counting elements is countably additive). For the counting function  $\mu(A) \stackrel{\text{def}}{=} |A|$  of Example 0.3, take any sequence of pairwise disjoint sets  $A_1, A_2, \dots$ . No two of them share an element, and  $D$  has only six elements, so at most six of the  $A_i$  contain any elements; the rest are  $\emptyset$ , with  $\mu(\emptyset) = 0$ . The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.3) shows that those terms add up to  $\mu$  of the union. So  $\mu$  is countably additive.

**Lemma 0.1** (Sums of non-negative terms). Let  $a_1, a_2, \dots$  be values in  $[0, \infty]$ , with partial sums  $s_n \stackrel{\text{def}}{=} \sum_{i=1}^n a_i$ , where  $x + \infty = \infty$  for every  $x$  in  $[0, \infty]$ . Then the partial sums are non-decreasing,  $s_1 \leq s_2 \leq \dots$ , so their limit, the sum  $\sum_{i=1}^{\infty} a_i = \lim_{n \rightarrow \infty} s_n$ , always exists, and it is either a finite number or  $\infty$ .

#### Proof

*Proof.* For each  $n$ :

$$\begin{aligned} s_{n+1} &= s_n + a_{n+1} && \text{(definition of } s_{n+1}) \\ &\geq s_n && (a_{n+1} \geq 0) \end{aligned}$$

If some partial sum  $s_N$  is  $\infty$ , then  $s_n = \infty$  for every  $n \geq N$ , so the limit is  $\infty$ . Otherwise, the partial sums form a non-decreasing sequence of real numbers. If that sequence is bounded above, it converges to a finite number (see [Wikipedia: Monotone convergence theorem](#)). If it is not bounded above, then for every number  $M$  some  $s_N$  exceeds  $M$ , and so does every later  $s_n \geq s_N$ ; the limit is  $\infty$ .  $\square$

*Remark.* By Lemma 0.1, the right-hand side of the equation in Definition 0.4 always has a value.

**Lemma 0.2** (Countable additivity and the empty set). If  $\mu$  is a *countably additive* function on a  $\sigma$ -algebra  $\mathcal{S}$ , then  $\mu(\emptyset) = 0$  or  $\mu(\emptyset) = \infty$ .

**i** Proof

*Proof.* The set  $\emptyset = S \setminus S$  is in  $\mathcal{S}$ , as the complement of  $S$ , and the sequence  $\emptyset, \emptyset, \dots$  is pairwise disjoint with union  $\emptyset$ , so:

$$\begin{aligned}\mu(\emptyset) &= \mu\left(\bigcup_{i=1}^{\infty} \emptyset\right) \quad (\text{the union of copies of } \emptyset \text{ is } \emptyset) \\ &= \sum_{i=1}^{\infty} \mu(\emptyset) \quad (\text{countable additivity})\end{aligned}$$

If  $\mu(\emptyset) = c$  for a finite  $c > 0$ , the right-hand side is  $c + c + \dots = \infty \neq c$ , a contradiction. So  $\mu(\emptyset)$  is 0 or  $\infty$ .  $\square$

**Example 0.5** (A countably additive function with  $\mu(\emptyset) = 0$ ). The counting function  $\mu(A) \stackrel{\text{def}}{=} |A|$  of Example 0.4 is countably additive, and:

$$\begin{aligned}\mu(\emptyset) &= |\emptyset| \quad (\text{definition of } \mu) \\ &= 0 \quad (\emptyset \text{ has no elements})\end{aligned}$$

**Example 0.6** (A countably additive function with  $\mu(\emptyset) = \infty$ ). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  (sets of die rolls), let  $\mu(A) \stackrel{\text{def}}{=} \infty$  for every set  $A$  in the  $\sigma$ -algebra of all subsets of  $D$  (Example 0.1), including  $A = \emptyset$ . For any sequence of pairwise disjoint sets  $A_1, A_2, \dots$ , the left-hand side of the countable additivity equation is:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \infty \quad (\text{definition of } \mu)$$

and the right-hand side, the limit of its partial sums (Lemma 0.1), is:

$$\begin{aligned}\sum_{i=1}^{\infty} \mu(A_i) &= \infty + \infty + \dots \quad (\text{definition of } \mu) \\ &= \infty \quad (\text{every partial sum is } \infty)\end{aligned}$$

The two sides agree, so  $\mu$  is countably additive, and  $\mu(\emptyset) = \infty$ .

*Remark.* Example 0.5 and Example 0.6 show that both values allowed by Lemma 0.2 occur. So  $\mu(\emptyset) = 0$  is an extra requirement, not a consequence of countable additivity.

**Theorem 0.3** (Countable additivity implies finite additivity). *If  $\mu$  is a countably additive function on a  $\sigma$ -algebra  $\mathcal{S}$ , and  $\mu(\emptyset) = 0$ , then  $\mu$  is finitely additive.*

**i** Proof

*Proof.* Let  $A_1, \dots, A_n$  be pairwise disjoint sets in  $\mathcal{S}$ , and extend them to a sequence by setting  $A_{n+1} = A_{n+2} = \dots = \emptyset$ . The extended sequence is still pairwise disjoint, since  $\emptyset$  shares no element with any set, and its union is  $A_1 \cup \dots \cup A_n$ . So:

$$\begin{aligned}
 \mu(A_1 \cup \dots \cup A_n) &= \mu\left(\bigcup_{i=1}^{\infty} A_i\right) && (A_i = \emptyset \text{ for } i > n) \\
 &= \sum_{i=1}^{\infty} \mu(A_i) && (\text{countable additivity}) \\
 &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) && (A_i = \emptyset \text{ for } i > n) \\
 &= \sum_{i=1}^n \mu(A_i) && (\mu(\emptyset) = 0)
 \end{aligned}$$

□

**Example 0.7** (Finitely additive but not countably additive). Let  $S = \{0, 1, 2, \dots\}$ , with the  $\sigma$ -algebra of all subsets of  $S$  (Theorem 0.2), and define:

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

$\mu$  is *finitely additive*. Let  $A_1, \dots, A_n$  be pairwise disjoint subsets of  $S$ . If every  $A_i$  is finite, then so is their union, and:

$$\begin{aligned}
 \mu(A_1 \cup \dots \cup A_n) &= 0 && (\text{a finite union of finite sets is finite}) \\
 &= \sum_{i=1}^n 0 && (\text{a sum of zeros is } 0) \\
 &= \sum_{i=1}^n \mu(A_i) && (\text{each } A_i \text{ is finite})
 \end{aligned}$$

If some  $A_k$  is infinite, then the union, which contains  $A_k$ , is infinite too, and:

$$\begin{aligned}
 \mu(A_1 \cup \dots \cup A_n) &= \infty && (\text{the union is infinite}) \\
 &= \mu(A_k) + \sum_{i \neq k} \mu(A_i) && (\mu(A_k) = \infty, \text{ and } \infty + x = \infty) \\
 &= \sum_{i=1}^n \mu(A_i) && (\text{regroup the terms})
 \end{aligned}$$

$\mu$  is *not countably additive*. The single-element sets  $\{0\}, \{1\}, \{2\}, \dots$  are pairwise disjoint, and their union is  $S$ , which is infinite. So:

$$\begin{aligned}\mu\left(\bigcup_{k=0}^{\infty}\{k\}\right) &= \mu(S) \quad (\text{the union is } S) \\ &= \infty \quad (S \text{ is infinite})\end{aligned}$$

but:

$$\begin{aligned}\sum_{k=0}^{\infty} \mu(\{k\}) &= 0 + 0 + \dots \quad (\text{each } \{k\} \text{ is finite}) \\ &= 0 \quad (\text{every partial sum is } 0)\end{aligned}$$

*Remark.* Example 0.7 shows that the converse of Theorem 0.3 fails, even when  $\mu(\emptyset) = 0$ : a finitely additive function need not be countably additive (see [Wikipedia: Sigma-additive set function — An additive function which is not  \$\sigma\$ -additive](#)).

## Measures

**Definition 0.5** (Measure). A **measure** on a set  $S$  with a  $\sigma$ -algebra  $\mathcal{S}$  is a function  $\mu : \mathcal{S} \rightarrow [0, \infty]$  that satisfies:

- $\mu(\emptyset) = 0$ .
- $\mu$  is **countably additive**.

**Example 0.8** (Counting elements is a measure). The counting function  $\mu(A) \stackrel{\text{def}}{=} |A|$  of Example 0.3, defined on the  $\sigma$ -algebra of all subsets of  $D$  (Example 0.1), takes values in  $[0, \infty]$ , is countably additive (Example 0.4), and gives  $\mu(\emptyset) = 0$ , so it is a measure.

**Corollary 0.1** (Measures are finitely additive). *Every **measure** is **finitely additive**.*

### Proof

*Proof.* A measure  $\mu$  is countably additive and has  $\mu(\emptyset) = 0$  (Definition 0.5), so Theorem 0.3 applies to it. □

**Definition 0.6** (Counting measure). The **counting measure** on a set  $S$  is the **measure** on the  $\sigma$ -algebra of all subsets of  $S$  that assigns each finite set its number of elements, and each infinite set the value  $\infty$ :

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

**Example 0.9** (Counting measure on the non-negative integers). For the counting measure  $\mu$  on  $\{0, 1, 2, \dots\}$ ,  $\mu(\{0, 1, 2\}) = 3$ , and the set of even numbers has  $\mu(\{0, 2, 4, \dots\}) = \infty$ . The even numbers are the union of the pairwise disjoint sets  $\{0\}, \{2\}, \{4\}, \dots$ , and countable additivity agrees:  $\sum_{k=0}^{\infty} \mu(\{2k\}) = 1 + 1 + \dots = \infty$ .

*Remark.* A [measure](#) generalizes size: it can measure how many elements a set has, as in Definition 0.6, or how long a set of real numbers is, as [Lebesgue measure](#) does, assigning each interval  $[a, b]$  its length  $b - a$ . Both appear as reference measures in the [Fubini–Tonelli theorem](#).